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Course: Deep Learning(MDM)

Batch : T3

Assignment 2: Implementing Explainable AI Techniques in Machine Learning and CNN Models

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Explainable AI Techniques on Titanic Dataset

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1. Introduction

Explainable Artificial Intelligence (XAI) plays a crucial role in modern machine learning by improving transparency and interpretability of models. Traditional machine learning models often function as black boxes, making it difficult to understand how predictions are generated. This lack of transparency can reduce trust, especially in critical applications.

The objective of this project is to implement a machine learning model and apply XAI techniques to interpret its predictions. The Titanic dataset is used as a case study, where the goal is to predict passenger survival. Techniques such as SHAP and LIME are applied to analyze both global and local model behavior.

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2. Dataset Description

The Titanic dataset contains information about passengers, including demographic and socio-economic attributes. The dataset consists of 891 records and multiple features such as:

- Age: Passenger age
- Sex: Gender
- Pclass: Passenger class (1st, 2nd, 3rd)
- Fare: Ticket price
- Embarked: Port of embarkation
- SibSp and Parch: Family-related features

The target variable is **Survived**, which indicates whether a passenger survived (1) or not (0).

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3. Data Preprocessing

Several preprocessing steps were performed to prepare the dataset for model training:

- Missing values in the Age column were filled using the median to reduce the effect of outliers
- Missing values in Embarked were filled using the mode
- The Cabin column was removed due to excessive missing values
- Irrelevant features such as Name, Ticket, and PassengerId were dropped
- Categorical variables were converted into numerical format using one-hot encoding

The dataset was then split into training and testing sets using an 80-20 ratio.

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4. Exploratory Data Analysis

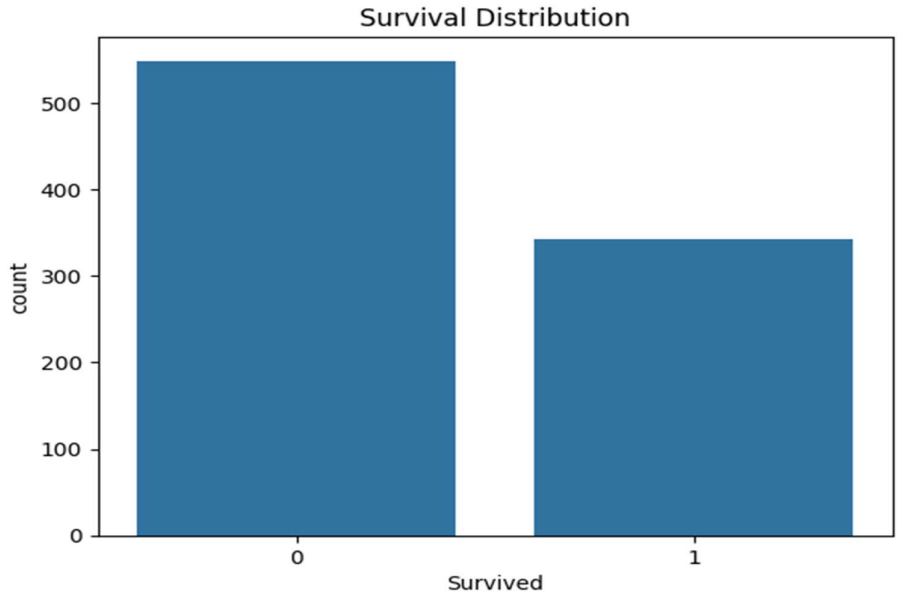
Exploratory data analysis was conducted to understand patterns in the dataset.

The survival distribution shows a slight class imbalance, with more passengers not surviving.

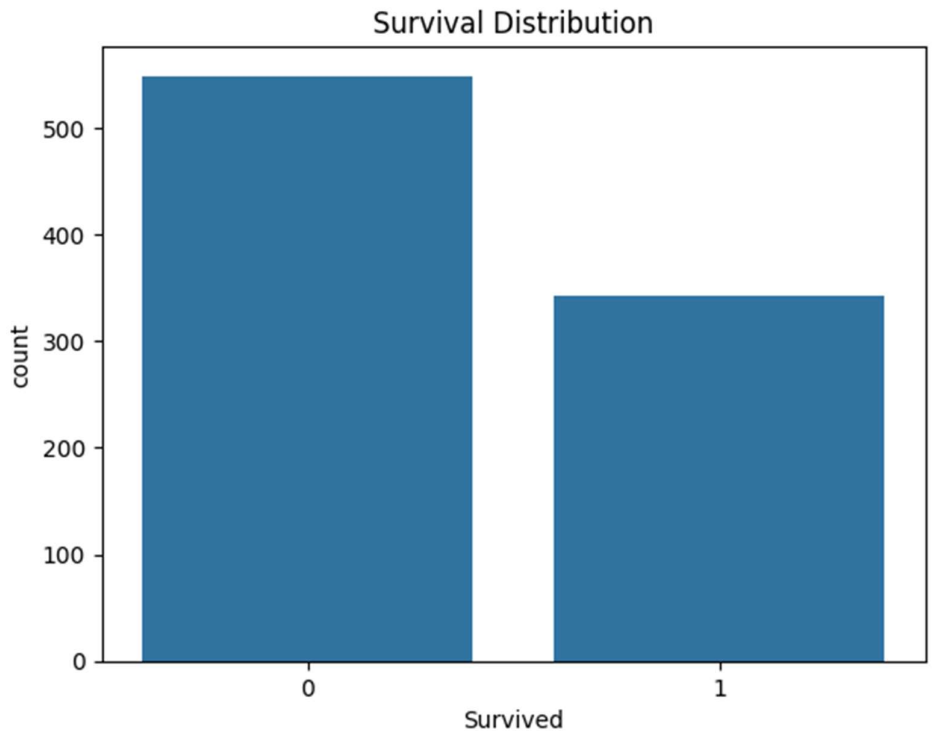
Gender-based analysis reveals that female passengers had a significantly higher survival rate compared to male passengers.

Passenger class analysis shows that individuals in higher classes had better survival chances, indicating the influence of socio-economic status.

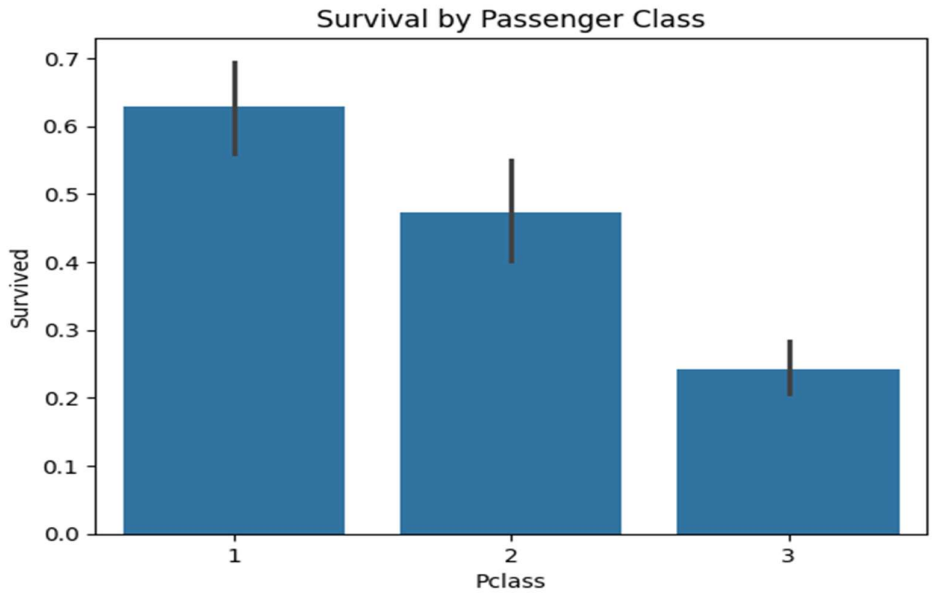
Survival Distribution Plot



Gender vs Survival Plot



Class vs Survival Plot



5. Model Implementation

A Random Forest Classifier was used for prediction. This model was chosen because it is robust, handles non-linear relationships, and provides feature importance.

The model was trained on the training dataset and evaluated using accuracy, classification report, and confusion matrix.

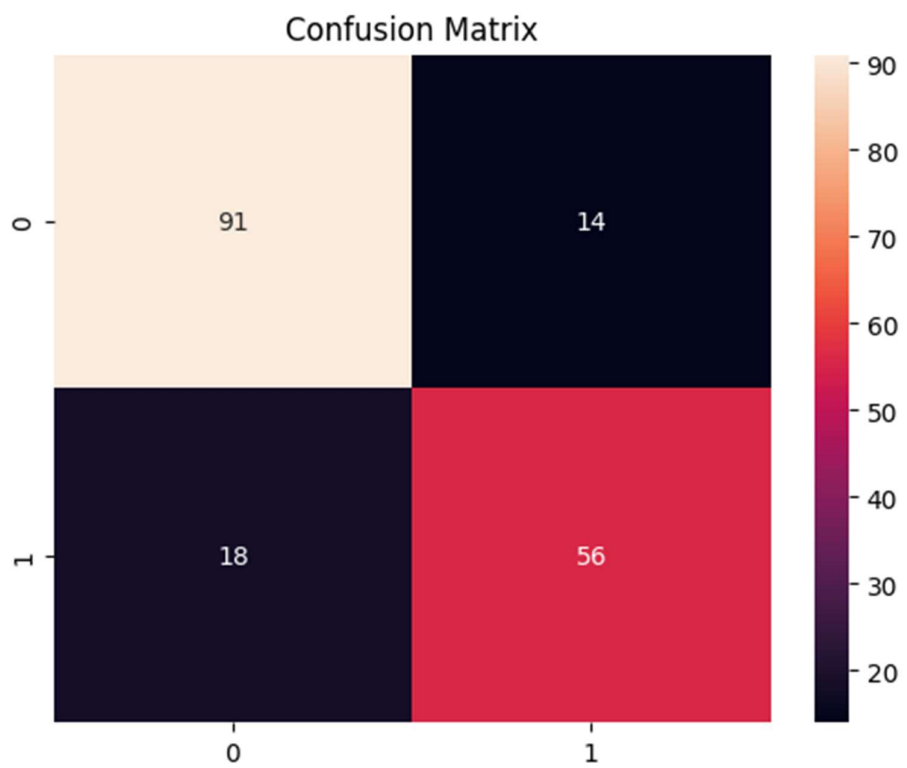
The results indicate that the model performs well with balanced precision and recall. However, some misclassifications are observed due to overlapping feature patterns.

```
•••  Accuracy: 0.8212290502793296
                  precision    recall  f1-score   support

               0       0.83      0.87      0.85       105
               1       0.80      0.76      0.78        74

        accuracy                           0.82       179
       macro avg       0.82      0.81      0.81       179
    weighted avg       0.82      0.82      0.82       179
```

Confusion Matrix



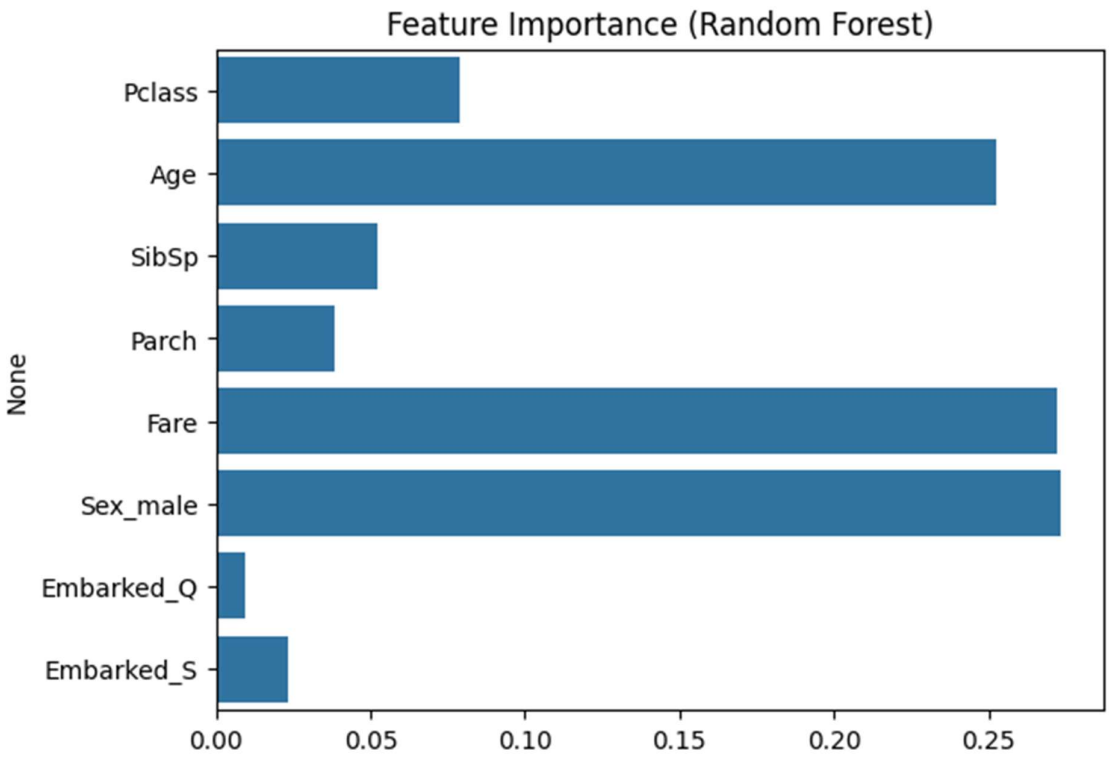
6. Explainable AI Techniques

To improve interpretability, multiple XAI techniques were applied.

6.1 Feature Importance

Feature importance from the Random Forest model shows that features such as Sex, Fare, and Pclass are the most influential.

Feature Importance Plot



6.2 SHAP (SHapley Additive exPlanations)

SHAP provides a global understanding of model behavior.

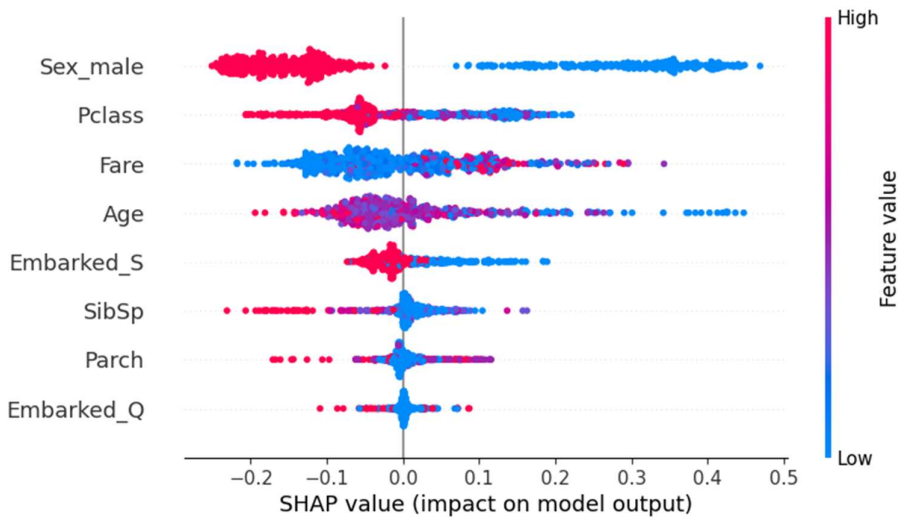
The SHAP summary plot shows that:

- Higher fare values positively influence survival
- Male gender negatively impacts survival
- Passenger class plays a significant role

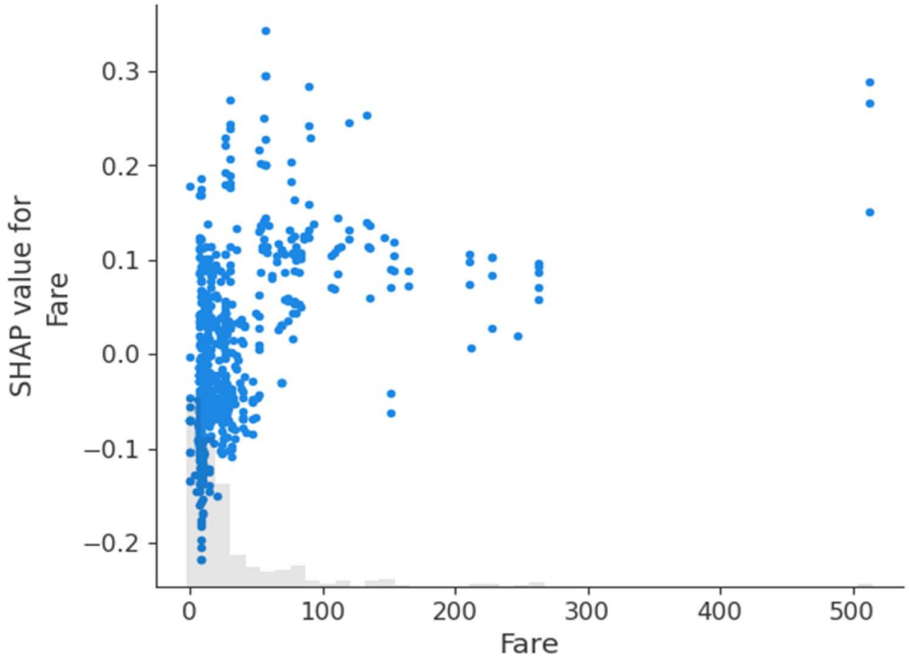
The SHAP scatter plot further shows a positive relationship between fare and survival probability.

The SHAP waterfall plot explains individual predictions by showing how each feature contributes to the final output.

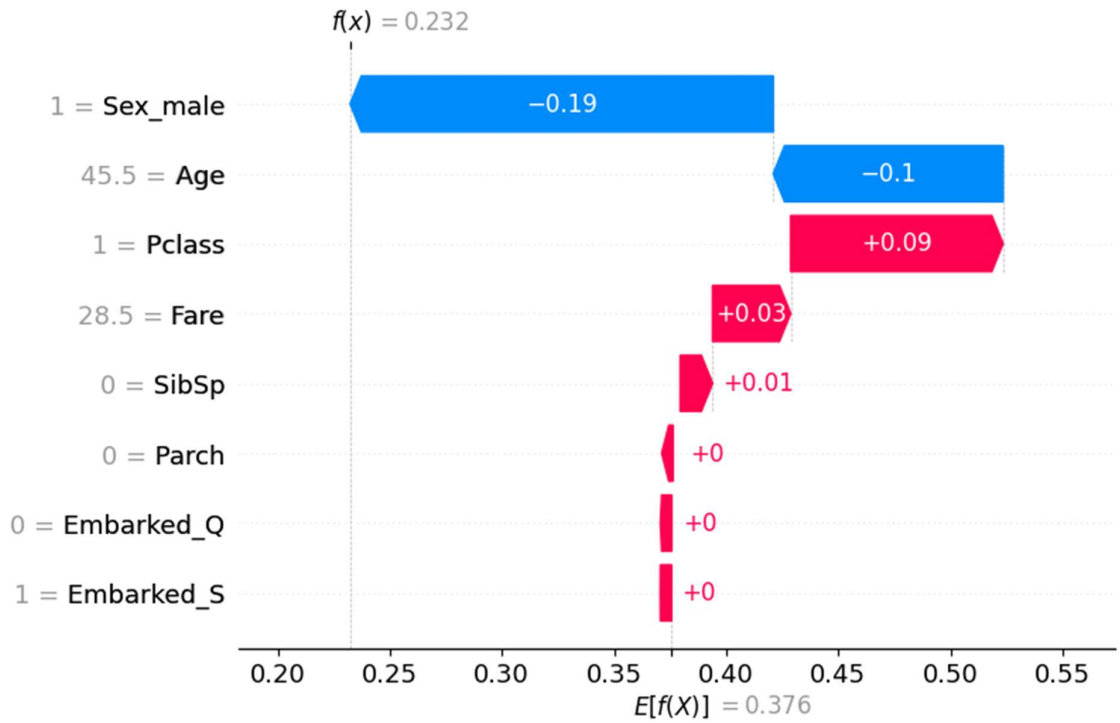
SHAP Summary (Beeswarm Plot)



SHAP Scatter Plot (Fare)



SHAP Waterfall Plot

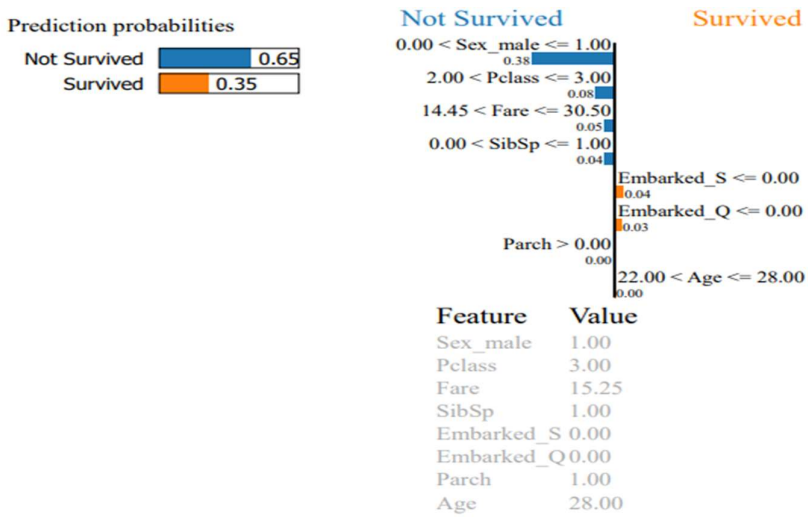


6.3 LIME (Local Interpretable Model-Agnostic Explanations)

LIME provides local explanations by approximating the model behavior for individual instances.

It highlights the most important features contributing to a specific prediction and makes the model decisions more interpretable.

LIME Explanation Output

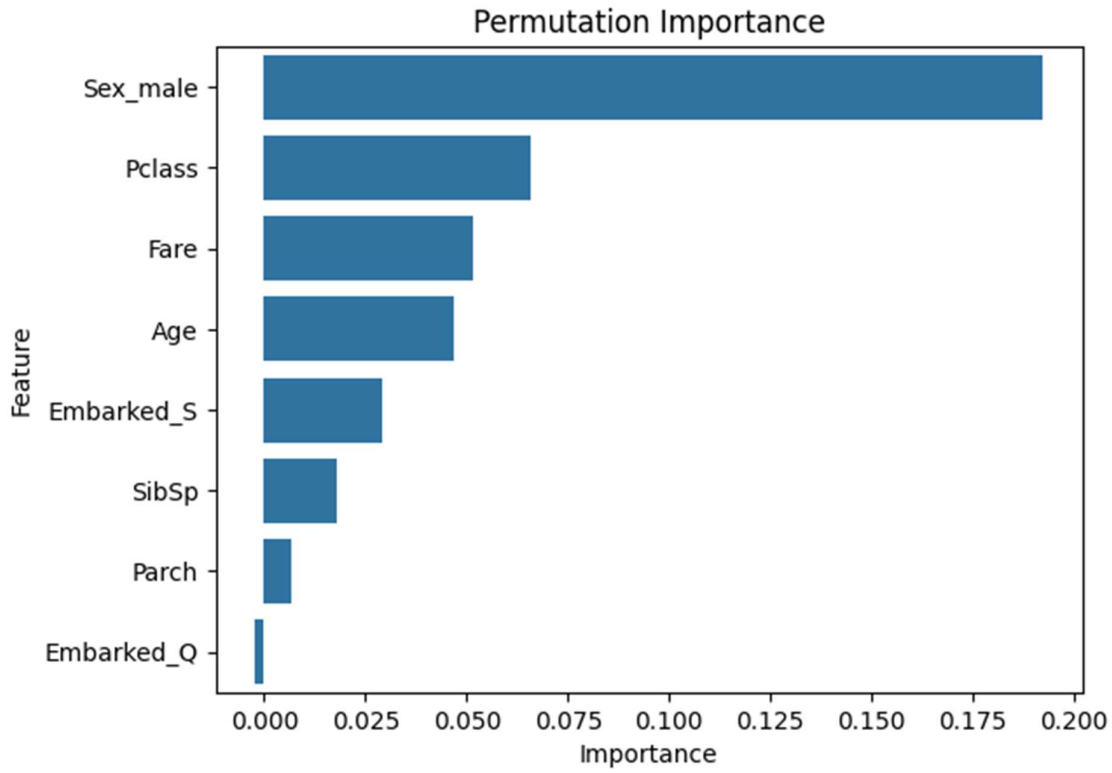


6.4 Permutation Importance

Permutation importance measures how model performance changes when feature values are shuffled.

The results confirm that Sex, Fare, and Pclass are the most important features, validating the findings from SHAP and feature importance.

Permutation Importance Plot

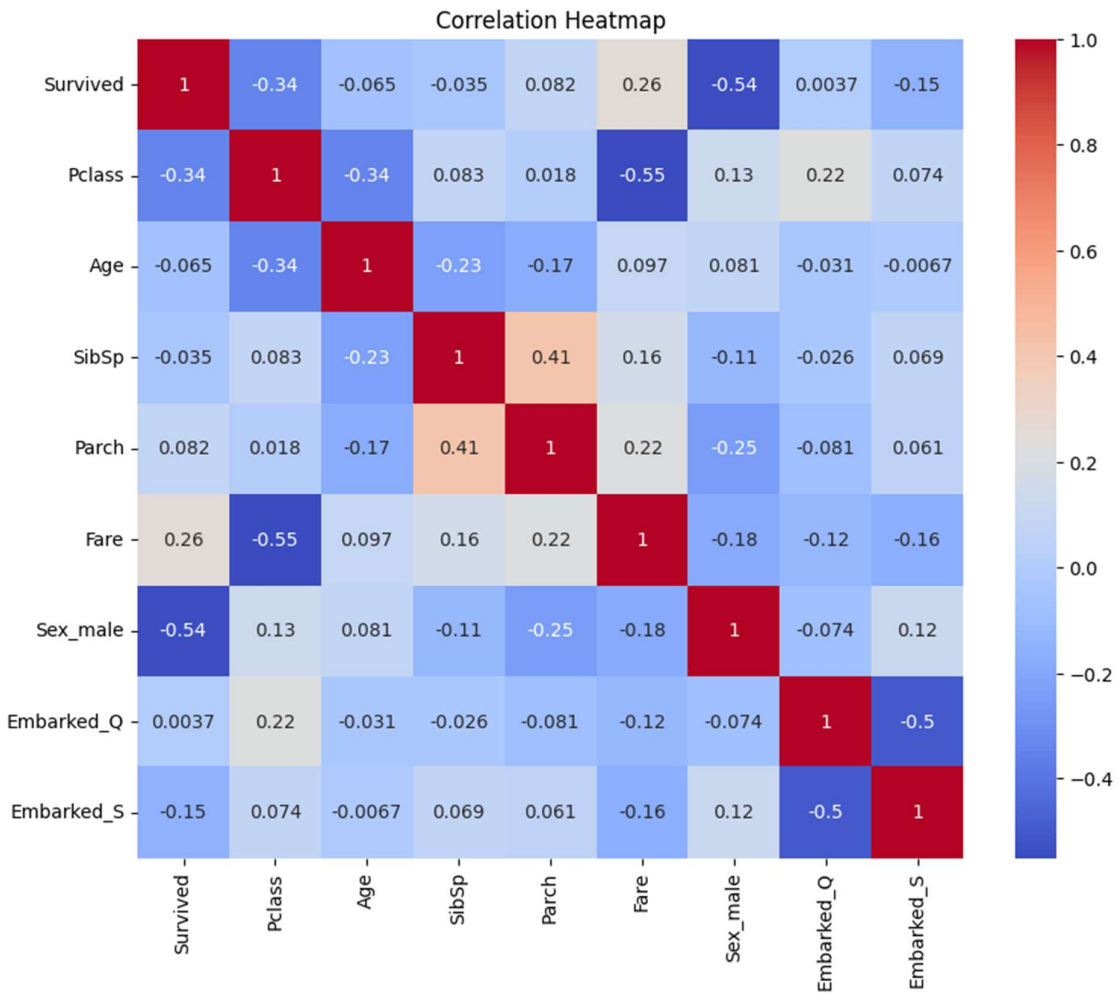


7. Visualization and Analysis

A correlation heatmap was generated to understand relationships between features.

The heatmap shows no strong multicollinearity, indicating that the features are relatively independent and suitable for modeling.

Correlation Heatmap



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8. Comparison of XAI Techniques

SHAP provides both global and local explanations by quantifying feature contributions across the dataset. LIME focuses on local interpretability by explaining individual predictions.

Both methods show consistent results, identifying gender, fare, and passenger class as the most influential features. This consistency increases confidence in the model.

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9. Bias and Fairness Analysis

The model shows a strong dependence on gender and passenger class. Female passengers and higher-class individuals are more likely to be predicted as survivors.

While this reflects historical patterns in the dataset, it introduces bias and raises fairness concerns in real-world applications. Explainable AI helps identify such biases and improves transparency.

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10. Limitations

- The model is trained on historical data and may not generalize well
- SHAP computations can be computationally expensive
- LIME provides approximate explanations
- The dataset itself contains inherent bias

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11. Conclusion

This project demonstrates how Explainable AI techniques can enhance transparency and interpretability in machine learning models.

The Random Forest model effectively predicts survival, and XAI techniques such as SHAP and LIME provide valuable insights into model behavior. The analysis reveals that gender, fare, and passenger class are the most influential features.

Explainable AI improves trust, helps identify bias, and enables better understanding of model decisions.

Github Repo: https://github.com/shivraj-nalawade/Explainable-AI-Techniques-in-Machine-Learning-.git